14. Write a code using RAW Sockets to implement Packet Sniffing.

Aim:

To write a code using RAW Sockets to implement Packet Sniffing.

```
From scapy.all import Sniff
From scapy.layers.inet import IP, TCP, UDP
                              ICMP
def Packet_callback (Packet):
    if IP in Packet:
        IP_layer = Packet [IP]
        Protocol = IP_layer.Proto
        Src_ip = IP._layer.src
        dst. ip = IP _layer.dst

        # Determine the Protocol
        Protocol_name = ""
        if Protocol == 1:
            Protocol_name = "ICMP"
        elif Protocol == 6:
        Protocol_name = "TCP"
        elif Protocol = 17:
            Protocol_name = "UDP"
        else:
        Protocol_name = "Unknown Protocol"
        # Print Packet details
        Print (f"protocol: {Protocol_name}")
```

```
Print(f"Source IP: {src_ip}")
Print(f"Destination IP: {dst_ip}")
Print("."*50)

#Capture Packets on the default netw
interface
Sniff(iface='Wi-Fi', prn=Packet_Ca
      filter="ip", store=0)
```

Output

```
Protocol: TCP
Source IP: 57.155.141.117
Destination IP: 192.168.1.6

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Result:-

Thus the python program for packet sniffer using TCP/UDP is performed.